*Proposed Solution to #5824 SSMJ*

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**Statement of the Problem:** Calculate:

$$I := \int_2^3 \left( \frac{\ln(x+1)\ln(x)}{x-1} + \frac{\ln(x)\ln(x-1)}{x+1} + \frac{\ln(x+1)\ln(x-1)}{x} \right) dx$$

**Solution of the Problem:**
We rewrite the problem as:

$$= \int_2^3 \ln(x) \left( \frac{\ln(x+1)}{x-1} + \frac{\ln(x-1)}{x+1} \right) dx + \int_2^3 \frac{\ln(x+1)\ln(x-1)}{x} dx$$

$$= \int_2^3 \ln(x) \frac{d}{dx} \left[ \ln(x+1)\ln(x-1) \right] dx + \int_2^3 \frac{\ln(x+1)\ln(x-1)}{x} dx$$

Using integration by parts on the first integral,

$$= \ln(x)\ln(x+1)\ln(x-1)\Big|_2^3 - \int_2^3 \frac{1}{x}\ln(x+1)\ln(x-1) dx + \int_2^3 \frac{\ln(x+1)\ln(x-1)}{x} dx$$

Notice the elegant cancellation of two integrals,

$$= \ln(3)\ln(4)\ln(2) - \ln(2)\ln(3)\ln(1) = \ln(2)\ln(3)\ln(4)$$

which is the required answer.